Name : Kanika Saini

SAP ID: 590034098

Program: Btech CSE (B16)

HDOC Day-11 – C Programming

Problem Statement

Write a **menu-driven C program using `switch-case`** to input a positive integer and perform one of the following operations:

1. Find the **sum and product of the first and last digits**.
   Example: `12345` → Sum = `6`, Product = `5`
2. Count how many digits are **lesser than, equal to, and greater than** a given digit `k`.
   Example: `12345`, `k = 3` → Lesser = `2`, Equal = `1`, Greater = `2`

---

1. Algorithm

Step 1: Start

Step 2:

Declare integer variables:

```
num, temp, first, last, digit, choice, k, lesser, equal, greater
```

Step 3:

Input a positive integer `num`.

Step 4:

Display the menu:

```
1. Find sum and product of first and last digits
2. Count digits lesser than, equal to, and greater than k
```

Step 5:

Input the user's `choice`.

Step 6:

Use `switch(choice)` to perform the selected operation.

Case 1 — Sum and Product of First and Last Digits

1. Find the last digit:

```
last = num % 10
```

2. Store the number in `temp`:

```
temp = num
```

3. Repeatedly divide `temp` by `10` until only the first digit remains:

```
while(temp >= 10)
temp = temp / 10
```

4. Store the first digit:

```
first = temp
```

5. Calculate:

```
Sum = first + last

Product = first * last
```

6. Display the first digit, last digit, sum, and product.

Case 2 — Compare Each Digit with `k`

1. Input the digit `k`.
2. Initialize:

```
lesser = 0
equal = 0
greater = 0
```

3. Store `num` in `temp`.
4. Extract each digit:

```
digit = temp % 10
```

5. Compare `digit` with `k`:
   - If `digit < k`, increment `lesser`.
   - If `digit == k`, increment `equal`.
   - Otherwise, increment `greater`.
6. Remove the processed digit:

```
temp = temp / 10
```

7. Repeat until all digits have been checked.
8. Display `lesser`, `equal`, and `greater`.

Step 7:

For any other menu choice, display **"Invalid choice"**.

Step 8: Stop.

---

2. Test Case Table

Test Case	Number	Choice	k	Expected Output	Actual Output	Result
1	12345	1	—	Sum = 6, Product = 5	Sum = 6, Product = 5	Pass
2	56789	1	—	Sum = 14, Product = 45	Sum = 14, Product = 45	Pass
3	12345	2	3	Lesser = 2, Equal = 1, Greater = 2	Lesser = 2, Equal = 1, Greater = 2	Pass
4	11234	2	1	Lesser = 0, Equal = 2, Greater = 3	Lesser = 0, Equal = 2, Greater = 3	Pass
5	98765	2	7	Lesser = 2, Equal = 1, Greater = 2	Lesser = 2, Equal = 1, Greater = 2	Pass

---

3. C Program

```
#include <stdio.h>

int main()
{
```

```
int num, temp, first, last, digit;
int choice, k;
int lesser = 0, equal = 0, greater = 0;

printf("Enter a positive integer: ");
scanf("%d", &num);

printf("\nMENU\n");
printf("1. Find sum and product of first and last digits\n");
printf("2. Count digits lesser than, equal to, and greater than k\n");

printf("Enter your choice: ");
scanf("%d", &choice);

switch(choice)
{
    case 1:
        last = num % 10;
        temp = num;

        while(temp >= 10)
        {
            temp = temp / 10;
        }

        first = temp;

        printf("First digit = %d\n", first);
        printf("Last digit = %d\n", last);
        printf("Sum = %d\n", first + last);
        printf("Product = %d\n", first * last);

        break;

    case 2:
        printf("Enter digit k: ");
        scanf("%d", &k);

        temp = num;
        lesser = 0;
        equal = 0;
        greater = 0;

        while(temp > 0)
        {
            digit = temp % 10;

            if(digit < k)
            {
                lesser++;
            }
            else if(digit == k)
            {
                equal++;
            }
            else
            {
```

```
                    greater++;
                }

                temp = temp / 10;
            }

            printf("Count_lesser = %d\n", lesser);
            printf("Count_equal = %d\n", equal);
            printf("Count_greater = %d\n", greater);

            break;

        default:
            printf("Invalid choice.\n");
    }

    return 0;
}
```

---

4. Compile and Execute

Save the program as:

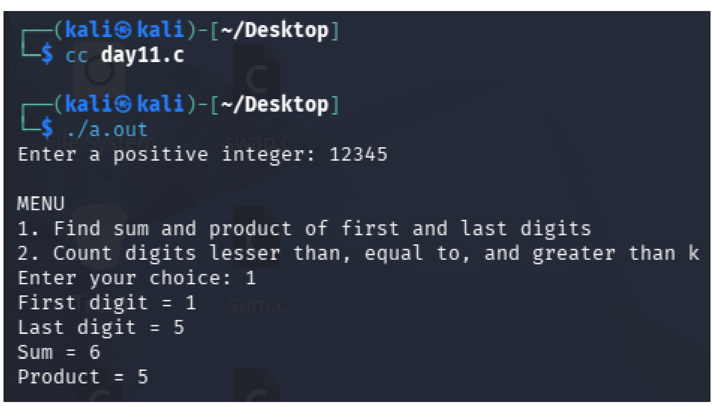
```
day11.c
```

Compile using the **`cc` compiler**:

```
cc day11.c
```

Run the program:

```
./a.out
```



```
┌──(kali㉿kali)-[~/Desktop]
└─$ ./a.out
Enter a positive integer: 56789

MENU
1. Find sum and product of first and last digits
2. Count digits lesser than, equal to, and greater than k
Enter your choice: 1
First digit = 5
Last digit = 9
Sum = 14
Product = 45
```

```
┌──(kali㉿kali)-[~/Desktop]
└─$ ./a.out
Enter a positive integer: 12345

MENU
1. Find sum and product of first and last digits
2. Count digits lesser than, equal to, and greater than k
Enter your choice: 2
Enter digit k: 3
Count_lesser = 2
Count_equal = 1
Count_greater = 2
```

```
┌──(kali㉿kali)-[~/Desktop]
└─$ ./a.out
Enter a positive integer: 11234

MENU
1. Find sum and product of first and last digits
2. Count digits lesser than, equal to, and greater than k
Enter your choice: 2
Enter digit k: 1
Count_lesser = 0
Count_equal = 2
Count_greater = 3
```

```
┌──(kali㉿kali)-[~/Desktop]
└─$ ./a.out
Enter a positive integer: 98765

MENU
1. Find sum and product of first and last digits
2. Count digits lesser than, equal to, and greater than k
Enter your choice: 2
Enter digit k: 7
Count_lesser = 2
Count_equal = 1
Count_greater = 2
```

---

5. Testing Against Prepared Use Cases

Test Case 1 — First and Last Digits

Input:

```
Enter a positive integer: 12345

MENU
1. Find sum and product of first and last digits
2. Count digits lesser than, equal to, and greater than k
Enter your choice: 1
```

First digit = `1`
Last digit = `5`

Therefore:

```
Sum = 1 + 5 = 6
```

```
Product = 1 × 5 = 5
```

Expected Output: Sum = `6`, Product = `5`
Actual Output: Sum = `6`, Product = `5`
Result: PASS ✓

Test Case 2 — Compare Digits with k

Input:

```
Enter a positive integer: 12345

MENU
1. Find sum and product of first and last digits
2. Count digits lesser than, equal to, and greater than k
Enter your choice: 2

Enter digit k: 3
```

Compare every digit with `k = 3`:

- `1 < 3` → Lesser
- `2 < 3` → Lesser
- `3 = 3` → Equal
- `4 > 3` → Greater
- `5 > 3` → Greater

Therefore:

```
Count_lesser = 2
Count_equal = 1
Count_greater = 2
```

Expected Output: `2, 1, 2`
Actual Output: `2, 1, 2`
Result: PASS ✓

Test Case 3 — Repeated Digits

Input: `11234`
Choice: `2`
k: `1`

Comparison:

```
1 = 1, 1 = 1, 2 > 1, 3 > 1, 4 > 1
```

Therefore:

```
Count_lesser = 0
Count_equal = 2
Count_greater = 3
```

Expected Output: `0, 2, 3`
Actual Output: `0, 2, 3`
Result: PASS ✓

Final Result

The menu-driven C program was successfully **written, compiled, executed, and tested**. The `switch-case`, `while` loop, and `if-else` conditions correctly perform both required operations, and all prepared test cases **PASS**.